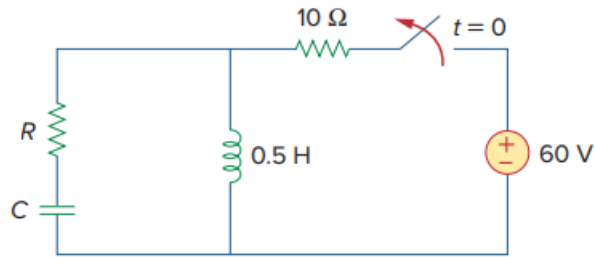


Problem

The switch in Fig. 8.91 is opened at $t = 0$ after the circuit has reached steady state. Choose R and C such that $\alpha = 8$ Np/s and $\omega_d = 30$ rad/s.

Figure 8.91:



Step-by-step solution

Step 1 of 2

For $t > 0$, we have a source free series RLC circuit.

We know that $\alpha = \frac{R}{2L}$

$$R = 2\alpha L$$

$$R = 2 \times 8 \times 0.5$$

$$\boxed{R = 8\Omega}$$

Step 2 of 2

We have $\omega_d = \sqrt{\omega_o^2 - \alpha^2}$

$$\omega_o = \sqrt{\omega_d^2 + \alpha^2}$$

$$\omega_o = \sqrt{30^2 + 8^2}$$

$$\omega_o = 31.048$$

$$\omega_o = \frac{1}{\sqrt{LC}}$$

$$C = \frac{1}{\omega_o^2 L}$$

$$C = \frac{1}{(31.048)^2 \times 0.5}$$

$$\boxed{C = 2.075\text{mF}}$$